Multiplayer Blackjack

Software Requirements Specification

Revision History

Date	Revision	Description	Author
02/15/2026	1.0	Initial Version	Michael, Curtis, Emma
02/25/2026	2.0	Started requirement list (2.3), added modules, filled out first three sections of SRS	Curtis, Michael, Emma
02/26/2026	2.5	Filled functional requirements (3.1)	Emma
02/27/2026	2.6	Added to functional requirements (3.1)	Emma
02/28/2026	2.7	Modified dependencies (2.5), modified and added to functional requirements (3.1), created non-functional requirements (4), resolved a contradiction from high-level features and player module requirements (2.3)	Curtis
03/01/2026	2.8	Added pages for references, modified 2.3, 3.1.7, 3.1.2, and 3.1.3	Curtis, Emma
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1. Purpose

This document outlines the requirements for the Multiplayer Blackjack program.

1.1. Scope

This document will catalog the user, system, and hardware requirements for the Multiplayer Blackjack program. It will not, however, document how these requirements will be implemented.

1.2. Definitions, Acronyms, Abbreviations

Term	Definition
Balance	How much currency a player user can use to bet with
Bust	When the cards total surpasses 21
Shoe	A set of cards composed of multiple decks shuffled together
Hand	The cards a player possesses in a round
Hit	To request an additional card to add to the current hand
Hole card	The first card drawn by the dealer, which is dealt face down
Rank	The numerical point value of a card
Stand	To keep the current hand during a turn

1.3. References

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UML Use Case Diagrams – Pages X-X

UML Class Diagrams – Pages X-X

UML Sequence Diagrams – Pages X-X

1.4. Overview

The Multiplayer Blackjack program simulates the experience of walking into a casino and playing blackjack. Players join a table with a dealer and bet from their balance. Players can hit or stand to get their score as close to 21 as possible without going over. If a player's score is better than the dealer's, they win their bets, but if the dealer has a better score, the player loses their bets.

2. Overall Description

2.1. Product Perspective

The Multiplayer Blackjack program is perfect for casinos that wish to provide their users with a mobile experience for this classic card game.

2.2. Product Architecture

The system will be organized into 6 major modules: The Server Module, The Client Module, The Dealer Module, The Player Module, The Game Module, and The Table Module.

2.3. Product Functionality/Features

The high-level features of the system are as follows (see section 3 of this document for more detailed requirements that address these features):

2.3.1. Users are split into two roles: dealers and players
2.3.2. Client connects to server
2.3.3. User logs into the client application with username and password
2.3.4. The system will have a main menu
2.3.5. From the main menu, the user can quit, join a table, or check the balance
2.3.6. Players can deposit or withdraw from their balance
2.3.7. Each table can take 6 players and 1 dealer
2.3.8. The Blackjack game starts when a table has at least 1 player and 1 host
2.3.9. Players choose what they bet each game under a time limit
2.3.10. Bets are validated
2.3.11. All users draw two cards, starting with the dealer, going clockwise, one at a time
2.3.12. The turn order goes counterclockwise, starting with the player closest to the dealer
2.3.13. Users can see every users' cards, excluding the dealer's hole card
2.3.14. Users can hit or stand during their turn
2.3.15. Players who stand cannot take another turn
2.3.16. Users have a time limit on their turns
2.3.17. Users draw from the shoe, which consists of six decks of standard playing cards
2.3.18. Users whose ranks add to over 21 bust, with players losing bets and the dealer causing the whole table to win theirs
2.3.19. Dealers must stand at least 17
2.3.20. After all players stand or bust, the dealer hits until they must stand or bust
2.3.21. Users whose ranks are below the dealer's loses their bets
2.3.22. Users whose ranks are above the dealer's win their bets
2.3.23. Users whose ranks match the dealer's keep their bets
2.3.24. Blackjacks are an instant win for users, prioritizing dealers
2.3.25. Aces count as either 11 or, if 11 would cause a bust, 1 in terms of rank
2.3.26. Face cards all count as 10 in terms of rank
2.3.27. Players can leave tables at any time
2.3.28. Dealers can leave after a blackjack game finishes
2.3.29. Users who disconnect may be timed out from joining another table to prevent players not losing their bets
2.3.30. Users who join mid-game can only start playing when the next game starts
2.3.31. Games start every 10 seconds at a table if none are in progress
2.3.32. Servers should keep a record of players' balances, usernames, passwords, and roles
2.3.33. Players/Dealer have a time limit to make choices

2.4. Constraints

2.4.1. The program must be programmed in Java
2.4.2. The program must use TCP/IP for communication
2.4.3. The program must have a GUI.
2.4.4. The program must consist of a server application and client application.
2.4.5. The program should have no web component.
2.4.6. No databases should be used for saving information

2.5. Assumptions and Dependencies

2.5.1. The program is assumed to run on hardware that supports a mouse and keyboard for user input.

2.5.2. The program is assumed to run on hardware that tracks time.

2.5.3. The server is assumed to run on hardware that can support the connection of multiple clients.

2.5.4. A client is assumed to run on hardware that supports network connection.

3. Specific Requirements

3.1. Functional Requirements

3.1.1. Common Requirements:

3.1.1.1. There are two types of users of the system: dealer and player.
3.1.1.2. The system stores account data (username, password, balance).
3.1.1.3. Passwords are encrypted in storage.

3.1.2. Server Module Requirements:

3.1.2.1. The server allows multiple client connections simultaneously.

3.1.2.2. The server stores account data when users register.

3.1.2.3. The server validates users when they sign in.

3.1.2.4. The server stores users' balances.

3.1.2.5. The server picks a random table that has the least number of players for a player to join.

3.1.2.6. The server hosts tables and their blackjack games.

3.1.2.7. The server validates bets by checking users' balances.

3.1.2.8. The tracks bet amounts for the round.

3.1.2.9. The server keeps track of the shoe and the cards players hold and draw.

3.1.2.10. The server issues time outs to players who disconnect from the server or table mid-game to prevent players from keeping lost bets.

3.1.2.11. The server informs the client of any validation errors or time outs.

3.1.3. Client Module Requirements:

3.1.3.1. The client connects to the server on start up.

3.1.3.2. Users navigate the client application using a GUI.

3.1.3.3. The client can receive player balance from the server.

3.1.3.4. The client can send back a modified player balance to the server to be stored.

3.1.3.5. The client can request the server to find a table for the client to join.

3.1.3.6. During a blackjack round, the client receives information from the server on what cards a user is holding and what the dealer's non-hole card is.

3.1.3.7. The client allows the user to place a bet by selecting chips.

3.1.3.8. The client tells the server what a user has bet.

3.1.3.9. The client tells the server when a user has stood or hit.

3.1.3.10. When the client is informed of a validation error or time out, that information is displayed to the user.

3.1.4. Dealer Module Requirements:

3.1.4.1. New dealer accounts can be created.
3.1.4.2. Dealers sign into their account using username and password.

3.1.4.3. Failed dealer login alerts user and prompts again.

3.1.4.4. A dealer account can only be signed in on one device at a time.

3.1.4.5. At the beginning of a round, the dealer first deals two cards to each player (all face up), then two to themself (one face up and one face down).

3.1.4.6. After all players have played, the dealer plays following fixed rules.

3.1.4.7. The dealer must hit on a hand of 16 or less and stand on a hand of 17 or more.

3.1.4.8. The dealer must continue to hit until their card total adds to 17 or more.

3.1.4.9. Once the dealer finishes their turn, the round is completed.

3.1.4.10. The dealer reveals their hand at the end of the round.

3.1.5. Player Module Requirements

3.1.5.1. New player accounts can be created.

3.1.5.2. Players sign into their account using their username and password.

3.1.5.3. Failed player login alerts the user and prompts again.

3.1.5.4. A player account can only be signed in on one device at a time.

3.1.5.5. Players have a balance attached to their account.

3.1.5.6. Players can deposit into and withdraw from their balance. This is entered manually by the player.

3.1.5.7. The player must place a bet at the beginning of each round.

3.1.5.8. A player places a bet by selecting chips.

3.1.5.9. When a player places a bet, the bet amount is removed from the player's balance.

3.1.5.10. A player cannot bet more than what is available in their balance. Bets are validated.

3.1.5.11. During a round of blackjack, a player has a hand made up of cards.

3.1.5.12. A player starts every round with two cards that are visible to everyone.

3.1.5.13. A player’s turn times out after 3 minutes. A turn timeout is an automatic stand.

3.1.5.14. During a player's turn, they can either hit (get another card to add to their hand) or stand (stay with their current hand).

3.1.5.15. During a player’s turn, they can continue to hit until their card total surpasses 21, at which point they have busted.

3.1.5.16. Cards received with hits are visible to everyone.

3.1.5.17. A player's turn ends once they either stand or bust.

3.1.5.18. A bust is an automatic loss.

3.1.5.19. At the end of a round a player loses or gains from their bet depending on how their hand compares to the dealer's hand.

3.1.5.20. If the player wins, the winnings are added to the player’s balance.

3.1.6. Game Module Requirements:

3.1.6.1. A card deck contains 52 cards.

3.1.6.2. A card deck is made up of 4 suits and 13 ranks.

3.1.6.3. A card deck can be shuffled.

3.1.6.4. A shoe consists of 6 mixed card decks.

3.1.6.5. The goal of the game blackjack is to get the card total of your hand as close to 21 as possible without going over.

3.1.6.6. Each player plays individually against the dealer.

3.1.6.7. The turn order goes counterclockwise, starting with the player closest to the dealer

3.1.6.8. Number cards 2-10 count as their respective numbers.

3.1.6.9. Face cards (king, queen, jack) each count as 10.

3.1.6.10. Ace cards count as either a 1 or an 11 depending on what is best for the hand.

3.1.6.11. A Blackjack is when the first two cards in a hand are an ace and a 10-value card.

3.1.6.12. A hand busts when its total surpasses 21.

3.1.6.13. The blackjack round cycle is betting, initial dealing, player turns, dealer turn, then bet resolution.

3.1.6.14. Bets are placed with chips.

3.1.6.15. Chips are tokens that equate to either $1, $5, $25, $50, $100, or $500.

3.1.6.16. Players play first.

3.1.6.17. The dealer plays last.

3.1.6.18. At the end of a round, hands are compared and bets are settled.

3.1.6.19. When the player's hand is larger than the dealer's hand, the player wins 2x their bet (1:1 winning).

3.1.6.20. When the dealer busts, the player wins 2x their bet (1:1 winning).

3.1.6.21. When the player busts, the player loses their bet.

3.1.6.22. When the dealer’s hand is larger than the player’s hand, the player loses their bet.

3.1.6.23. When the player’s hand and the dealer’s hand are the same, the player gets their bet back.

3.1.6.24. When the player has a Blackjack and does not tie with the dealer, the player wins 2.5x their bet (3:2 winning).

3.1.7. Table Module Requirements:

3.1.7.1. A table consists of at most 1 dealer.

3.1.7.2. A table can host 0-6 players.

3.1.7.3. A round of blackjack at an empty table begins 10 seconds after a player and a dealer join the table.

3.1.7.4. A table hosts consecutive rounds while the table is not empty.

3.1.7.5. A player cannot join a full table (6 players).

3.1.7.6. A new table automatically opens when all current tables are full of either players or dealers.

3.1.7.7. A dealer cannot join a table with a dealer already present in it.

3.1.7.8. Joining a table, a player must wait until the next round to play.

3.1.7.9. Multiple tables can be active at once.

3.1.7.10. Blackjack rounds start every 10 seconds at a non-empty table if none are in progress.

3.1.7.11. Users can leave a table.

3.1.7.12. Players can leave a table at any time.

3.1.7.13. Dealers can only leave while no round is in play.

3.1.7.14. Dealers can queue their leave mid-game.

3.1.7.15. Dealers who queued their leave will leave immediately after a game finishes.

3.2. External Interface Requirements

3.2.1. There will be a graphical user interface.
3.2.2. There will be a sign in menu. From this menu a user can sign into an existing account, register as a player, or register as a dealer.
3.2.3. If the user chooses to sign in they are taken to a menu that asks for a username and password for an existing account.
3.2.4. If the user chooses to register they are taken to a menu that asks for a username and password to register a new account.
3.2.5. After players sign in, there will be a main player menu. From this menu players can view their account balance, add funds to their account, cash out their account, join a blackjack table, or log out.
3.2.6. Players will be able to see their current updated balance while playing a game.
3.2.7. At the beginning of a round, a player places a bet by selecting from chips displayed.
3.2.8. During a player's turn, the player will have a button to hit and a button to stand.
3.2.9. At the end of a round, a player's outcome is displayed (win, loss, tie).
3.2.10. After dealers sign in, there will be a main dealer menu. From this menu dealers can host a table or log out.
3.2.11. The system must run on hardware with networking abilities, keyboard, mouse, and display.
3.2.12. Depositing or withdrawing from the players' balance is handled by a banking service.

3.3. Internal Interface Requirements

3.3.1. Server and clients will communicate over TCP/IP.
3.3.2. User actions passed from client to server.
3.3.3. Clients cannot modify game data.
3.3.4. Balance modifications are concurrent.
3.3.5. No component shall directly modify another component's internal data.

4. Non-Functional Requirements

4.1. Security and Privacy Requirements

4.1.1. Passwords are encrypted in storage
4.1.2. User balance modifications and betting is validated

4.2. Environmental Requirements

4.2.1. No databases should be used for saving information
4.2.2. The client requires an active internet connection to communicate with the server
4.2.3. The clients and servers are required to have a Java Runtime Environment

4.3. Performance Requirements

4.3.1. Users should be able to use their allotted decision-making time within a minute
4.3.2. The server must process a user's input action, such as a hit, within a second
4.3.3. The server should be able to support up to 5 tables at a time

UML Use Case Specifications

UML Use Case Diagrams

UML Class Diagram

UML Sequence Diagrams

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